

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 30-2023: A 50-Year-Old Woman with Confusion

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PRESENTATION OF CASE

Dr. Grace R. Bobkowsky (Medicine): A 50-year-old woman was admitted to this hospital because of confusion.

The patient had been in her usual state of health until 5 months before the current presentation, when dysphagia developed. One month later, she was evaluated by her primary care physician. She reported having a sensation that pills and food were getting stuck in her throat when she swallowed. She was unable to take her medications, and her food intake had decreased to two eggs per day. She estimated that her weight had decreased by 5 kg during the previous month. She was referred for esophagogastroduodenoscopy (EGD), which reportedly revealed ulceration of the gastrojejunal junction from a Roux-en-Y gastric bypass performed 12 years earlier but otherwise normal mucosa. Examination of biopsy specimens obtained during the procedure reportedly revealed findings consistent with lymphocytic esophagitis. Treatment with swallowed fluticasone was recommended.

Three months before the current presentation, leg swelling developed, and the patient was again evaluated by her primary care physician. She had stopped using fluticasone after 1 week of treatment because she had not noticed any decrease in the dysphagia. Her weight had decreased by 3 kg since the evaluation 1 month earlier. Pitting edema was present in the legs. A diagnosis of chronic venous insufficiency was considered, and compression stockings were recommended.

Two months before the current presentation, leg swelling worsened, arm swelling developed, and the patient had trouble performing activities of daily living. She was admitted to another hospital. She reported persistent dysphagia and reduced food intake. Her weight had increased by 3 kg since the evaluation 1 month earlier. Pitting edema was present in the arms and legs. The blood albumin level was 2.2 g per deciliter (reference range, 3.5 to 4.8), and the zinc level was 0.36 μg per milliliter (reference range, 0.66 to 1.10). Blood levels of vitamin B₂, vitamin B₁₂, folate, and methylmalonate were normal; the blood level of N-terminal pro-B-type natriuretic peptide was normal for the patient's age. Urinalysis revealed trace ketones but was otherwise normal. The albumin:creatinine ratio in a random urine

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Table 1. Laboratory Data.*

Variable	Reference Range, Other Hospital	2 Mo before Current Presentation, Other Hospital	Reference Range, This Hospital†	1 Mo before Current Presentation, This Hospital	On Current Presentation, This Hospital
Hemoglobin (g/dl)	12.0–15.5	13.0	12.0–16.0	12.5	11.7
Hematocrit (%)	34.9–44.5	37.8	36.0–46.0	39.6	37.2
Mean corpuscular volume (fl)	80–100	99	80.0–100.0	109.0	104.0
Platelet count (per μ l)	135,000–400,000	415,000	150,000–400,000	320,000	341,000
White-cell count (per μ l)	4000–11,000	14,270	4500–11,000	11,700	17,160
Differential count (per μ l)					
Neutrophils	1800–7500	5510	1800–7700	5150	10,690
Lymphocytes	1000–4800	7650	1000–4800	5800	5740
Monocytes	10–900	870	200–1200	630	560
Eosinophils	0–400	0	0–900	80	40
Basophils	0–200	240	0–300	30	40
Immature granulocytes	—	—	0–100	80	90
Sodium (mmol/liter)	136–145	139	135–145	138	140
Potassium (mmol/liter)	3.5–5.2	2.9	3.4–5.0	3.9	4.5
Chloride (mmol/liter)	95–106	108	98–108	102	105
Carbon dioxide (mmol/liter)	20–31	20	23–32	20	16
Urea nitrogen (mg/dl)	9–23	8	8–25	6	4
Creatinine (mg/dl)	0.50–1.30	0.95	0.60–1.50	0.58	0.53
Glucose (mg/dl)	74–106	111	70–110	106	118
Anion gap (mmol/liter)	3–17	11	3–17	16	19
Lactic acid (mmol/liter)	—	—	0.5–2.0	1.5	3.3
Calcium (mg/dl)	8.7–10.4	7.5	8.5–10.5	7.4	7.7
Phosphorus (mg/dl)	2.4–5.1	3.1	2.6–4.5	2.8	3.8
Magnesium (mg/dl)	1.6–2.6	1.7	1.7–2.4	2.0	1.7
Alkaline phosphatase (U/liter)	27–116	194	30–100	235	200
Alanine aminotransferase (U/liter)	10–49	48	7–33	68	38
Aspartate aminotransferase (U/liter)	6–40	29	9–32	32	29
Total bilirubin (mg/dl)	0.0–1.0	1.2	0.0–1.0	0.7	1.3
Direct bilirubin (mg/dl)	0.0–0.4	0.7	0.0–0.4	0.4	0.6
International normalized ratio	0.9–1.1	1.0	0.9–1.1	1.1	1.5
Albumin (g/dl)	3.5–4.8	2.2	3.3–5.0	1.7	1.9
Total protein (g/dl)	5.7–8.2	5.3	6.0–8.3	4.6	4.3

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. They may therefore not be appropriate for all patients.

sample was 11.6 mg of albumin per gram of creatinine (reference value, <30.0). Other laboratory test results are shown in Table 1.

Abdominal ultrasonography reportedly revealed an echogenic liver, a finding consistent with fatty infiltration, as well as trace free fluid. Pitting edema of the arms and legs was attributed to hypoalbuminemia from limited oral intake, which was related to dysphagia from untreated lymphocytic esophagitis. The use of fluticasone was encouraged, and supplemental nutritional drinks were recommended, along with a multivitamin and zinc sulfate. On the fourth hospital day, the patient was discharged home.

One month before the current presentation, the patient was admitted to this hospital because of confusion and a urinary tract infection (UTI) with *Escherichia coli*. Confusion resolved with treatment of the UTI. Her weight had decreased by 2 kg since the admission 1 month earlier. The patient reported loose stools without blood or mucus, which had occurred for 1 month. Edema was present. The blood albumin level was 1.7 g per deciliter, the zinc level 0.3 μg per milliliter, the selenium level 34 ng per milliliter (reference range, 70 to 150), the vitamin A level 16.6 μg per deciliter (reference range, 32.5 to 78.0), and the vitamin C level 0.2 mg per deciliter (reference range, 0.4 to 2.0). Blood levels of vitamin B₁ and vitamin B₃ were normal. Other laboratory test results are shown in Table 1.

EGD showed ulceration of the gastrojejunal anastomosis but otherwise normal mucosa. Biopsy specimens of the esophagus, stomach, and jejunum showed normal mucosa. A biopsy specimen of the gastrojejunal anastomosis showed small-intestinal mucosa with erosion and non-specific morphologic features; periodic acid-Schiff staining was negative for microorganisms, Congo red staining was negative for amyloid, and immunohistochemical staining was negative for cytomegalovirus. Magnetic resonance enterography of the abdomen and pelvis, performed with and without the administration of intravenous contrast material, revealed no evidence of active inflammatory bowel disease. A high-protein diet was recommended, along with a multivitamin, zinc sulfate, and selenium. On the eighth hospital day, the patient was discharged home.

After discharge, the patient initially felt well. The edema abated, and her appetite improved. She began to have six small meals and three protein shakes per day. Two weeks after discharge, and 1 week before the current presentation, confusion developed, and the gait became unsteady; she began using a walker for ambulation. During the next week, confusion increased, and she fell while walking. Her family brought her to the emergency department of this hospital for evaluation.

On the patient's arrival in the emergency department, she was unable to answer questions about her history. Her family reported that she had had slowed and slurred speech for the past week and increasing difficulty with attention, including difficulty with holding a conversation. Other medical history included chronic back pain, bipolar disorder, seizure disorder, and ulcerative colitis. Medications included carbamazepine, ergocalciferol, fluticasone, gabapentin, mesalamine, methocarbamol, mirtazapine, a multivitamin, olanzapine, omeprazole, oxycodone-acetaminophen, selenium, sucralfate, thiamine, topiramate, and zinc sulfate. There were no known drug allergies. The patient was receiving disability benefits and did not work because of chronic back pain. She had smoked two packs of cigarettes per day for 35 years. She did not drink alcohol or use illicit drugs. Her mother had endometrial cancer, and her father had lung cancer.

On examination, the temporal temperature was 36.8°C, the blood pressure 89/52 mm Hg, the pulse 132 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 28.9. Her weight had decreased by 6 kg since the admission 1 month earlier. She appeared chronically ill and was lethargic but arose in response to voice. She moved her arms and legs but did not participate in a detailed neurologic examination. Pitting edema was present in the arms and legs. The white-cell count was 17,150 per microliter (reference range, 4500 to 11,000), with neutrophil predominance. The blood zinc level was 0.35 μg per milliliter, the copper level 0.28 μg per milliliter (reference

range, 0.75 to 1.45), and the vitamin B₆ level less than 2 µg per liter (reference range, 5 to 50). Urinalysis revealed trace ketones and 50 to 100 white cells per high-power field (reference value, <10). Other laboratory test results are shown in Table 1.

A chest radiograph was normal, without evidence of pneumonia or pulmonary edema. Urine, blood, and stool specimens were obtained for culture. Three liters of intravenous lactated Ringer's solution were administered, and the systolic blood pressure increased to 100 mm Hg. Three hours after the patient's arrival in the emergency department, she became more lethargic and had no response to voice; she reacted to being shaken but did not open her eyes or answer questions. Additional imaging studies were performed.

Dr. Theodore T. Pierce: Computed tomography (CT) of the head, performed without the administration of contrast material, showed no abnormalities. CT of the chest, abdomen, and pelvis (Fig. 1), performed after the administration of intravenous contrast material, revealed diffuse hepatic steatosis, as well as mucosal hyperenhancement and submucosal edema involving portions of the small and large bowel, with relative sparing of the distal jejunum, proximal ileum, descending colon, and sigmoid colon.

Dr. Bobkowski: Treatment with intravenous vancomycin, cefepime, and metronidazole was initiated. Six hours after the patient's arrival in the emergency department, the blood pressure decreased to 85/45 mm Hg, and the heart rate increased to 132 beats per minute. There was no response to painful stimuli. Myoclonic jerking movements were observed. Treatment with norepinephrine was initiated, and the patient was transferred to the intensive care unit. The trachea was intubated for airway protection, and sedation with propofol and dexmedetomidine was initiated. Treatment with vancomycin, cefepime, and metronidazole was continued; prescribed medications of carbamazepine, a multivitamin, omeprazole, selenium, sucralfate, thiamine, topiramate, and zinc sulfate were administered.

On the first hospital day, the blood pressure increased, and vasopressors and sedative medications were discontinued. The patient had no spontaneous interaction or speech production. She did not follow commands or grimace in response to noxious stimuli. The pupils were equal

and reactive to light. The corneal reflexes were intact, as were the cough and gag reflexes. There was no spontaneous movement of the arms or legs, but they withdrew in response to pain; the deep-tendon reflexes were normal. The results of CT of the head were normal. Sedation was not resumed, and serial neurologic examinations were performed.

On the second hospital day, leukocytosis had resolved, but the patient's mental status had not improved. Electroencephalography (EEG) with long-term monitoring revealed near-continuous generalized periodic discharges with triphasic morphologic features, often with superimposed rhythmic activity. The results of magnetic resonance imaging (MRI) of the head were normal, as were the results of magnetic resonance angiography of the head and neck. Lumbar puncture was performed, and cerebrospinal fluid (CSF) analysis showed normal protein levels and no nucleated cells. The urine culture grew *Klebsiella pneumoniae*. The blood and stool cultures were negative, as was a stool test for *Clostridioides difficile* and an examination of the stool for ova and parasites. Treatment with vancomycin, cefepime, and metronidazole was stopped, and treatment with ciprofloxacin was started.

On the third hospital day, the patient's mental status had not improved. EGD showed a small scar at the gastrojejunal anastomosis but otherwise normal mucosa. The results of flexible sigmoidoscopy were normal. Biopsy specimens of the jejunum and descending colon showed normal mucosa.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

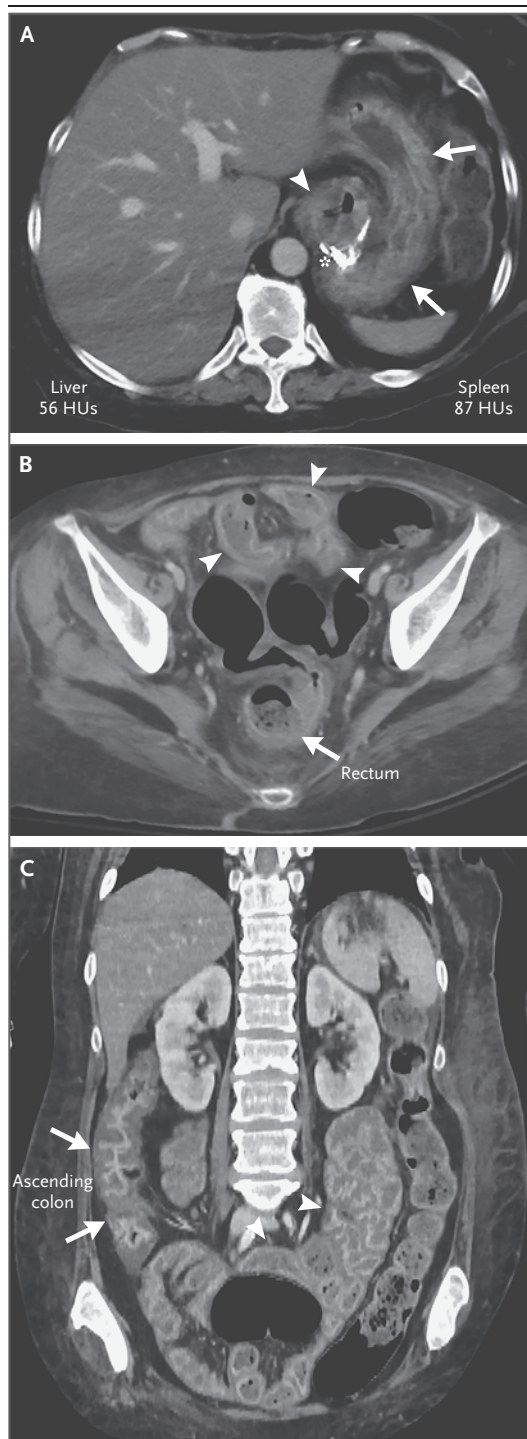
Dr. Amulya Nagarur: This 50-year-old woman with a history of Roux-en-Y gastric bypass, ulcerative colitis, bipolar disorder, and seizure disorder initially presented with malnutrition and anasarca and then had rapidly progressive encephalopathy. A review of the background, midground, and foreground of her presentation will inform the selection of "illness scripts" to consider in this case.

BACKGROUND

Five months before the current presentation, dysphagia developed, and the patient began to lose weight. Biopsy specimens obtained during

Figure 1. CT of the Abdomen and Pelvis.

CT of the abdomen and pelvis was performed after the administration of intravenous contrast material. An axial image of the upper abdomen (Panel A) shows findings that are typical in patients who have undergone Roux-en-Y gastric bypass: surgical clips (asterisk) separating the gastric pouch (arrowhead) from the excluded stomach (arrows). The homogeneous low density (measured in Hounsfield units [HUs]) of the liver, as compared with the spleen, reflects steatosis. An axial image of the pelvis (Panel B) and a coronal oblique reformation image (Panel C) show mild long-segment mural thickening and edema of the ascending colon and rectum (arrows), as well as mucosal hyperenhancement and minimal thickening of the jejunum and ileum (arrowheads). The presence of fat stranding in the body wall indicates edema.



EGD showed findings consistent with lymphocytic esophagitis, a disease entity that is known to cause dysphagia but is not completely understood.¹ Lymphocytic esophagitis is more common among older people and may be triggered by gastroesophageal reflux disease, which could have developed in this patient after gastric bypass surgery.

An attempt at treatment with swallowed fluticasone did not increase the patient's oral intake. The fact that weight loss occurred despite progressive leg edema is suggestive of muscle loss. Two months before the current presentation, the patient was admitted to the hospital because of anasarca. It was hypothesized that poor oral intake from lymphocytic esophagitis had led to protein-calorie malnutrition and hypoalbuminemia.

A salient feature of this case is hypoalbuminemia, which can result from either a decrease in protein production or an abnormal loss of protein.^{2,3} The average adult liver synthesizes approximately 15 g of albumin per day; the serum half-life of albumin is approximately 20 days. Synthetic dysfunction in the liver can lead to hypoalbuminemia. However, this patient had no other biochemical evidence suggestive of liver synthetic dysfunction. Congenital analbuminemia is an autosomal recessive disorder caused by defects in the gene that encodes albumin, but it is exceedingly rare. Nephrotic syndrome can cause an abnormal loss of albumin, but this possibility was ruled out with urine testing performed during the first hospital admission. Profound inflammation can lead to increased

vascular permeability and extravasation of albumin from the intravascular space to the extravascular space; such inflammation was not observed in this patient. Protein-losing enteropathy remained a possible diagnosis at that time.

MIDGROUND

One month before the current presentation, more clues emerged. The patient was found to have additional nutrient deficiencies and reported loose stools. Further investigations for protein-losing enteropathy, including EGD and magnetic resonance enterography, were unrevealing. At this point, protein-losing enteropathy seemed unlikely, and hypoalbuminemia was attributed to poor oral intake. It is important to recognize that the liver can increase production of albumin to compensate for a deficiency, which suggests that another process, in addition to poor oral intake, may have been contributing to hypoalbuminemia.

FOREGROUND

One week before the current admission, the patient had rapidly progressive encephalopathy and an unsteady gait that resulted in a fall. A UTI with *K. pneumoniae* was diagnosed. Hypotension abated after the administration of vasopressors and systemic antibiotic agents, but altered mental status persisted.

ALTERED MENTAL STATUS

The differential diagnosis for altered mental status in this patient is broad. One helpful diagnostic schema categorizes infectious and non-infectious diseases according to intracranial or extracranial involvement.

Intracranial infections — such as encephalitis, neurosyphilis, and acquired prion disease — are unlikely in this patient, given that she had normal results on CSF analysis. A UTI was diagnosed, but altered mental status worsened despite treatment with antibiotics. Other extracranial infections were ruled out with blood and stool cultures and additional stool studies; abdominal imaging showed no signs of intraabdominal infection.

Noninfectious intracranial processes, such as hemorrhagic or ischemic stroke, were ruled out with CT and MRI. This patient had an underlying seizure disorder, but EEG findings were consistent with a toxic–metabolic process, rather than seizure activity. Autoimmune or paraneoplastic encephalitis, vasculitis, and sporadic prion disease are all unlikely, given the normal results on CSF analysis. Rapidly progressive dementia remains a possible diagnosis. However, this patient had several risk factors that make a nonin-

fectious extracranial cause of altered mental status more likely.⁴

This patient was taking several medications that can cause altered mental status, including carbamazepine, gabapentin, methocarbamol, olanzapine, oxycodone–acetaminophen, and topiramate.⁵ However, none of these medications were newly prescribed. Although changes in her weight could have resulted in altered medication pharmacokinetics, the clinical presentation is not consistent with a class-specific toxidrome, and the altered mental status did not abate with time.

Could myxedema coma explain the altered mental status in this patient with progressive edema? She had pitting edema, whereas nonpitting edema would be expected in a patient with myxedema. In addition, there was no evidence of hypothermia, bradycardia, hypoventilation, hyponatremia, or hypoglycemia, features that would support a diagnosis of myxedema.⁶

In this patient with a history of Roux-en-Y gastric bypass and ulcerative colitis with recent malnutrition and hypoalbuminemia, nutritional deficiencies and metabolic encephalopathy are important noninfectious extracranial causes of altered mental status to consider.

NUTRITIONAL DEFICIENCIES

Nutritional deficiencies associated with ulcerative colitis include deficiencies in iron, vitamin B₁₂, fat-soluble vitamins, and zinc. Vitamin B₁₂ deficiency can cause altered mental status.⁷ However, this patient's ulcerative colitis did not appear to be active on magnetic resonance enterography, and the vitamin B₁₂ level was normal. Patients who have undergone Roux-en-Y gastric bypass may have macronutrient and micronutrient deficiencies that are related to poor oral intake and alterations in critical absorptive surfaces.⁸ This patient had several deficiencies associated with previous Roux-en-Y gastric bypass, including deficiencies in vitamin C, vitamin B₆, vitamin A, and trace minerals (zinc, copper, and selenium). However, the most common nutritional deficiencies that cause altered mental status after Roux-en-Y gastric bypass are deficiencies in vitamins B₁ and B₃, and these levels were normal in this patient. Although malnutrition is probably playing a role in this patient's presentation, the clinical picture does not fit a single nutrient deficiency syndrome.

HYPERAMMONEMIA

All tissues produce ammonia as a byproduct of amino acid metabolism. The primary route of ammonia excretion is through the urea cycle, with urinary excretion of ammonia as urea. Hyperammonemic encephalopathy can occur when perturbations in ammonia metabolism lead to an elevated ammonia level.⁹ Although an elevated ammonia level can be suggestive of this diagnosis in the presence of appropriate risk factors and clinical findings, testing for ammonia can notoriously yield falsely elevated results.^{10,11}

Hepatic encephalopathy is a clinical syndrome that can be encountered in patients with cirrhosis when portosystemic shunting causes ammonia to bypass the hepatic metabolism. Patients with hepatic encephalopathy can present with various manifestations of altered mental status and associated findings, such as asterixis. This patient had progressive altered mental status and myoclonic jerking but no history of cirrhosis. In addition, she had no findings on physical examination or laboratory test results that were suggestive of cirrhosis. However, she had several risk factors for noncirrhotic hyperammonemia.

NONCIRRHOTIC HYPERAMMONEMIA

There were no signs suggesting that this patient had hyperammonemia due to acute liver failure. Hyperammonemia can be caused by an increased protein load from an oral or parenteral source or from gastrointestinal bleeding. The patient had increased her protein intake substantially 1 week before the onset of altered mental status. Hyperammonemia can also be caused by a catabolic state, such as severe malnutrition.¹² The patient's weight loss despite the development of progressive edema was suggestive of muscle loss and sarcopenia. Muscle is an important extrahepatic site of ammonia metabolism, and severe sarcopenia can lead to hyperammonemia.¹³

Certain drugs and toxins, including antiepileptic agents such as valproate, can cause urea-cycle dysfunction and hyperammonemia. There are case reports of carbamazepine-induced asterixis with hyperammonemia, but it is exceedingly rare.^{14,15} Hyperammonemia can also be due to infections with urease-producing bacteria, including *K. pneumoniae*. Inherited and acquired urea-cycle disorders are considerations in this

patient, but her history of Roux-en-Y gastric bypass suggests the possibility of another clinical syndrome.

HYPERAMMONEMIA AFTER ROUX-EN-Y GASTRIC BYPASS

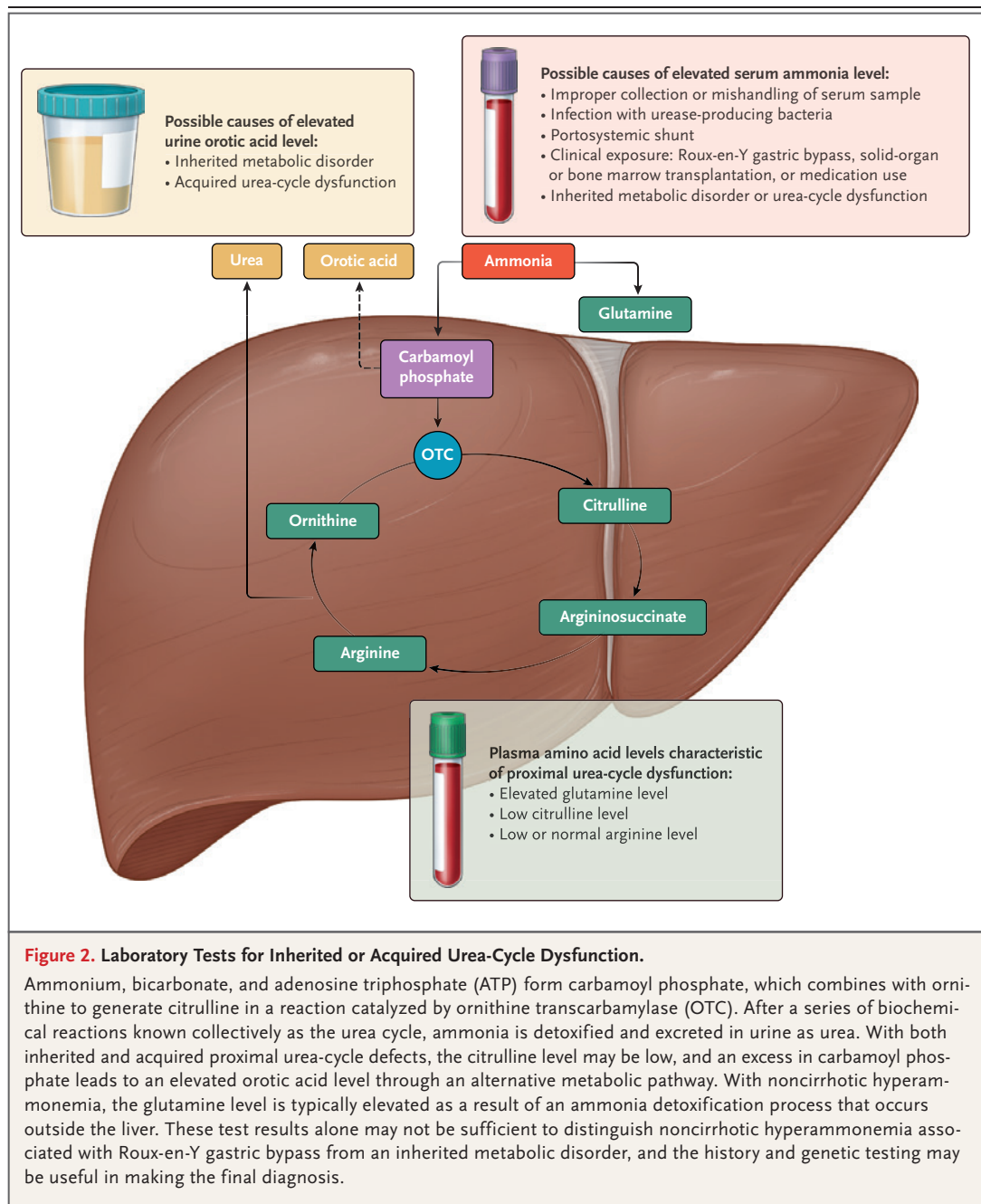
There is an increasingly recognized association between noncirrhotic hyperammonemia and a history of Roux-en-Y gastric bypass, and some experts have proposed naming this phenomenon the "gastric bypass hyperammonemia syndrome."¹⁶ Case reports describe patients with hyperammonemic encephalopathy who are typically middle-aged and female and do not have clinically significant liver disease. Hyperammonemic encephalopathy may develop at various points after gastric bypass surgery, with hallmark findings including hypoalbuminemia, nutritional deficiencies, an elevated plasma glutamine level, and a low zinc level. Delays in diagnosis and management may result in a poor prognosis with a high risk of death.¹⁶⁻¹⁸

Hyperammonemia associated with Roux-en-Y gastric bypass is more common in women than in men. It has been hypothesized that undiagnosed hemizygous pathogenic variants in the gene that encodes ornithine transcarbamylase (OTC) in women could contribute to an elevated risk of urea-cycle dysfunction. A role for zinc deficiency in urea-cycle enzymatic dysfunction has also been hypothesized. Increased production of ammonia, portosystemic shunting, an altered intestinal microbiome, and a profound catabolic state after Roux-en-Y gastric bypass may all contribute to the pathogenesis of this syndrome, although more investigation is needed to understand this rare and unique disease.^{17,19}

I suspect that this patient had noncirrhotic hyperammonemia after Roux-en-Y gastric bypass and that the diagnostic test was measurement of the serum ammonia level. The degree to which certain aspects of her presentation — including the high-protein diet, sarcopenia, metabolic effects of carbamazepine, and UTI with *K. pneumoniae* — contributed to her rapid decline is unclear. These are important considerations.

DR. AMULYA NAGARUR'S DIAGNOSIS

Noncirrhotic hyperammonemia after Roux-en-Y gastric bypass.



DIAGNOSTIC TESTING

Dr. Nina B. Gold: The diagnostic test was measurement of the serum ammonia level. The serum specimen must be obtained from a free-flowing venipuncture without the use of a tourniquet, stored on ice, and sent immediately to the laboratory; deviations from this process may lead to falsely elevated results. This patient's serum am-

monia level was 90 μmol per liter (reference range, 12 to 48) and remained elevated on repeat measurements, with a maximum level of 107 μmol per liter. She had encephalopathy during the time that these levels were obtained. The next step was an evaluation for causes of noncirrhotic hyperammonemia, including inherited metabolic disorders and acquired perturbations of the urea cycle.²⁰

The plasma amino acid levels obtained during the patient's hospitalization were low, a finding consistent with malnutrition. The free and total plasma carnitine levels, plasma acylcarnitine profile, serum creatine kinase level, and urine organic acid levels obtained after the hospitalization were normal. She was found to have a mildly elevated urine orotic acid level, which can be associated with proximal urea-cycle dysfunction. However, repeat measurements of the plasma amino acid levels have not shown hallmarks of a proximal urea-cycle defect, such as an elevated glutamine level or a low citrulline level (Fig. 2). Results of molecular testing for 76 monogenic disorders associated with hyperammonemia were normal, which makes a diagnosis of an inherited metabolic disorder, such as an ornithine transcarbamylase deficiency, unlikely.

The most likely reason for noncirrhotic hyperammonemia in this patient is her history of Roux-en-Y gastric bypass. Malnutrition, followed by the rapid initiation of a high-protein diet, might have led to acquired dysfunction of the urea cycle that resulted in noncirrhotic hyperammonemia.

CLINICAL AND LABORATORY DIAGNOSIS

Noncirrhotic hyperammonemia after Roux-en-Y gastric bypass.

DISCUSSION OF MANAGEMENT

Dr. Andrew Z. Fennes: Treatment for noncirrhotic hyperammonemia in this patient was focused on lowering the serum ammonia level and included the administration of oral rifaximin and lactulose. The use of a multivitamin and zinc and selenium supplements was continued, and treatment with an oral copper supplement was initiated. Renal replacement therapy was considered, but after 1 week of medical therapy, the ammo-

nia level normalized and the patient's mental status returned to normal. A surgical consultation was obtained regarding the need for gastric bypass reversal, given the risk of recurrent hyperammonemia and the high mortality reported with gastric bypass hyperammonemia syndrome.¹⁶

Dr. Elan R. Witkowski: This patient had a rare and potentially fatal complication of gastric bypass. Although the diagnosis of hyperammonemia is extremely uncommon, the overall scenario is not unusual. Patients with complications of Roux-en-Y gastric bypass can present with vague and atypical symptoms, such as nausea, rash, or neurologic manifestations. These symptoms can be benign and unrelated to surgery, but serious micronutrient deficiencies or metabolic complications must also be considered. Complications of bypass reversal can also occur, so surgery is often deferred until nonoperative management strategies have been exhausted.

Dr. Fennes: Three months after diagnosis, hypoalbuminemia and hyperammonemia recurred in this patient despite treatment with rifaximin and lactulose.

Dr. Witkowski: A gastrostomy tube was placed, and nutritional support was provided with tube feeds. The albumin and ammonia levels returned to normal. However, 2 years later, hypoalbuminemia and hyperammonemia again recurred. At that point, reversal of gastric bypass was performed. Since then, the patient has needed several endoscopic dilations of her gastrogastric anastomosis but otherwise has done well.

FINAL DIAGNOSIS

Noncirrhotic hyperammonemia after Roux-en-Y gastric bypass.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

REFERENCES

1. Pittman ME. Lymphocytic esophagitis: current understanding and controversy. *Am J Surg Pathol* 2022;46(1):e55-e63.
2. Rothschild MA, Oratz M, Schreiber SS. Serum albumin. *Hepatology* 1988;8:385-401.
3. Levitt DG, Levitt MD. Human serum albumin homeostasis: a new look at the roles of synthesis, catabolism, renal and gastrointestinal excretion, and the clinical value of serum albumin measurements. *Int J Gen Med* 2016;9:229-55.
4. Mattison MLP. Delirium. *Ann Intern Med* 2020;173(7):ITC49-ITC64.
5. Francis J. Drug-induced delirium. *CNS Drugs* 1996;5:103-14.
6. Kwaku MP, Burman KD. Myxedema coma. *J Intensive Care Med* 2007;22:224-31.
7. Radziszewska M, Smarkusz-Zarzecka J, Ostrowska L, Pogodziński D. Nutrition and supplementation in ulcerative colitis. *Nutrients* 2022;14:2469.
8. Bloomberg RD, Fleishman A, Nalle JE, Herron DM, Kini S. Nutritional deficiencies following bariatric surgery: what

- have we learned? *Obes Surg* 2005;15:145-54.
9. Auron A, Brophy PD. Hyperammonemia in review: pathophysiology, diagnosis, and treatment. *Pediatr Nephrol* 2012; 27:207-22.
 10. Wijdicks EFM. Hepatic encephalopathy. *N Engl J Med* 2016;375:1660-70.
 11. Barsotti RJ. Measurement of ammonia in blood. *J Pediatr* 2001;138:Suppl: S11-S20.
 12. Strong A, Gold J, Gold NB, Yudkoff M. Hepatic manifestations of urea cycle disorders. *Clin Liver Dis (Hoboken)* 2021; 18:198-203.
 13. Jindal A, Jagdish RK. Sarcopenia: ammonia metabolism and hepatic encephalopathy. *Clin Mol Hepatol* 2019;25:270-9.
 14. Rivelli M, el-Mallakh RS, Nelson WH. Carbamazepine-associated asterixis and hyperammonemia. *Am J Psychiatry* 1988; 145:269-70.
 15. Coulter DL, Allen RJ. Secondary hyperammonaemia: a possible mechanism for valproate encephalopathy. *Lancet* 1980; 1:1310-1.
 16. Fenves AZ, Shchelochkov OA, Mehta A. Hyperammonemic syndrome after Roux-en-Y gastric bypass. *Obesity (Silver Spring)* 2015;23:746-9.
 17. Nagarur A, Fenves AZ. Late presentation of fatal hyperammonemic encephalopathy after Roux-en-Y gastric bypass. *Proc (Bayl Univ Med Cent)* 2017;30:41-3.
 18. Acharya G, Mehra S, Patel R, Frunza-Stefan S, Kaur H. Fatal nonhepatic hyperammonemia in ICU setting: a rare but serious complication following bariatric surgery. *Case Rep Crit Care* 2016;2016: 8531591.
 19. Aquilio E, Spagnoli R, Riggio D, Seri S. Effects of zinc on hepatic ornithine transcarbamylase (OTC) activity. *J Trace Elem Electrolytes Health Dis* 1993;7: 240-1.
 20. Stergachis AB, Mogensen KM, Khoury CC, et al. A retrospective study of adult patients with noncirrhotic hyperammonemia. *J Inher Metab Dis* 2020;43: 1165-72.

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